

Final Project

Retail Sales Analysis

Project Details	Information
Project Title	Retail Sales Analysis
Database	Capstone
Tools Used	MS Excel, My Sql,Python,Power BI,MS Word
Project Type	Final Project
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Date	September 2026

Executive Summary & Table of Contents

Executive Summary

This project presents an end-to-end retail data analytics solution. The primary goal is to extract actionable business insights from raw transactional data using four analytical tools: Microsoft Excel, SQL, Python, and Microsoft Power BI. The pipeline includes data cleaning, relational database querying, RFM customer segmentation using machine learning, and interactive dashboard creation.

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Project Overview & Objectives

Problem Statement

In a competitive retail businesses need to track sales performance, understand customer behavior, and identify growth opportunities. This project aims to build an end-to-end analytics solution that enables insights generation from raw transactional data using a multi-tool approach.

Key Business Objectives

- Analyze revenue trends across multiple retail store branches and staff members.
- Monitor inventory levels and generate automated alerts for low stock.
- Segment customer records using behavioral metrics (Recency, Frequency, Monetary).
- Design an executive Power BI dashboard for real-time decision support.

Relational Data Schema & System Architecture

Dataset Schema

The dataset consists of 9 relational tables:

- customers: Profile and location details.
- orders: Transaction headers, tracking dates, store ID, and staff ID.
- order_items: Line-item details including quantity, list price, and discounts.
- products: Master catalog with category ID and brand ID.
- brands & categories: Product classification metadata.
- stocks: Product quantities per store.
- stores & staffs: Store locations and employee hierarchy.

Technical Workflow

Excel (Cleaning) → SQL (Storage & Querying) → Python (EDA & ML)
→ Power BI (Dashboarding) → MS Word(Report)

Phase 1 – Excel: Data Standardization & Cleaning

Column Standardization & Duplicates Removal

- Standardized all column names across 9 workbooks into consistent snake_case format (e.g., customer_id, order_date).
- Applied Excel's "Remove Duplicates" tool across primary key columns in customers and order_items to eliminate duplicate entries.

Missing Value Handling

- Identified missing phone numbers and ZIP codes using conditional formatting and blank filters.
- Replaced missing text entries with standard "NA" placeholders to preserve row integrity during SQL ingestion.

Excel: Data Validation & Feature Engineering

Data Type Casting & Validation

- Converted transaction dates (order_date, required_date, shipped_date) from text to standardized Excel Date format (YYYY-MM-DD).
- Applied strict Data Validation dropdown lists for order_status with allowed inputs: *Pending*, *Rejected*, *Shipped*, and *Delivered*.

Derived Columns Calculation

Added a calculated column in order_items to compute net revenue per item:

$$\text{total_price} = (\text{list_price} \times \text{quantity}) - \text{discount}$$

Excel: Outlier Analysis & CSV Export

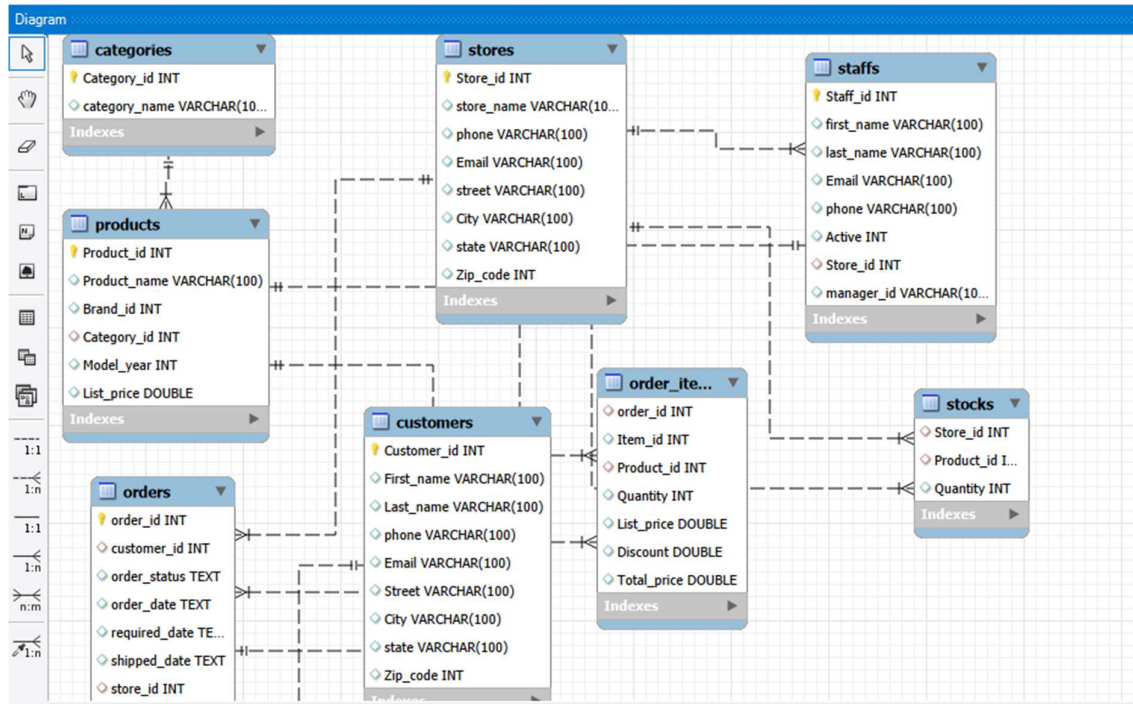
Lookup Merging & Pivot Summaries

- Utilized XLOOKUP and VLOOKUP to bring product_name and category_id directly into transaction sheets for validation.
- Generated summary Pivot Tables to calculate sales distributions and spot extreme pricing anomalies.

Phase 2 – SQL: Table Schema & Data Loading

Relational Table Creation (DDL)

Defined normalized tables in MySQL with explicit primary keys, foreign keys, and cascading constraints matching the project ER diagram.



SQL: Aggregations & Business Performance

Store-Wise Sales Query

Calculated total gross revenue grouped by individual retail branch stores

```
select s.store_name, round(sum(oi.total_price), 2) as total_revenue
```

```
from capstone.orders as o
```

```
join capstone.order_items as oi
```

```
on o.order_id = oi.order_id
```

```
join capstone.stores as s
```

```
on o.store_id = s.store_id
```

```
group by s.store_name
```

```
order by total_revenue desc;
```

SQL: Inventory Monitoring & Segmentation Table

Low Stock Alert Query

Identified inventory items requiring reordering where total units fall below the minimum threshold:

```
select str.store_id,p.product_id,p.Product_name,s.quantity
from capstone.stores as str
join capstone.stocks as s
on s.Store_id = str.Store_id
join capstone.products as p
on p.Product_id = s.Product_id
where s.quantity < 10;
```

Customer Segmentation Staging Table

Created a destination table to receive processed RFM machine learning labels from Python:

```
create table capstone.customer_segments(customer_id varchar(50),recency
int,frequency int,monetary decimal(12,2),segment varchar(50));
```

Phase 3 – Python: Database Connection & EDA

SQL Data Extraction

Extracted relational records into Pandas DataFrames using SQLAlchemy and mysql.connector

```
import pandas as pd
```

```
from sqlalchemy import create_engine
```

```
engine =
```

```
create_engine('mysql+mysqlconnector://root:password@localhost/retail_db')
```

Exploratory Data Analysis (EDA)

- Inspected summary statistics using `df.describe()` and `df.info()`.
- Verified numerical distributions, sales spread, and detected skewness using Seaborn boxplots and histograms.

Python: RFM Feature Engineering

Metrics Calculation

Engineered three core customer transaction features:

- **Recency (R):** Number of days between snapshot date and latest order date per customer.
- **Frequency (F):** Distinct count of orders placed by each customer.
- **Monetary (M):** Sum of `total_price` across all customer purchases.

Python: Customer Segmentation & Export

Clustering and Label Assignment

- Normalized RFM metrics and categorized customers into three behavioral clusters: **Loyal**, **New**, and **At Risk**.
- High frequency with recent purchases were tagged as *Loyal*, recent single purchases as *New*, and high dormancy as *At Risk*.

Exporting to SQL Database

Appended segmented labels into the `customer_segments` MySQL table using `.to_sql()`

Phase 4 – Power BI: Data Modeling & Navigation

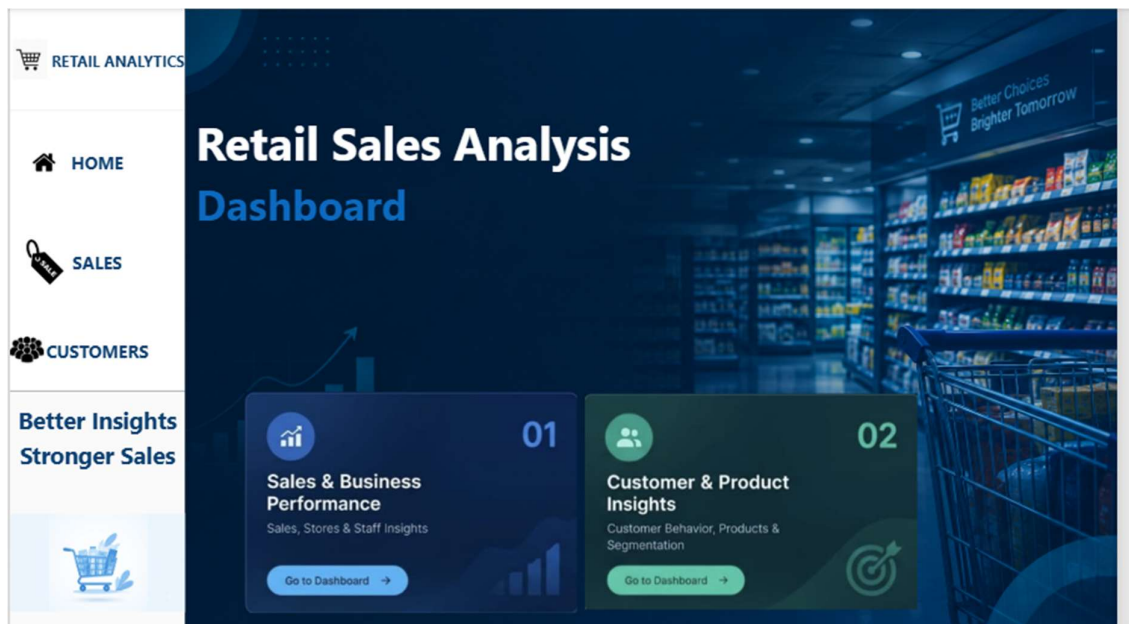
Star Schema Data Modeling

- Imported all core business tables (orders, order_items, products, customers, stores, staffs, customer_segments) into Power BI Desktop via MySQL connection.
- Built one-to-many relationships in Model View connecting Dimension tables to the central order_items fact table.

Dashboard Navigation System

Implemented a dark-themed UI with a sidebar navigation pane:

- **HOME:** Welcome landing page featuring project badges.
- **SALES:** Direct link to Sales & Business Performance.
- **CUSTOMERS:** Direct link to Customer & Product Insights.



SALES & BUSINESS PERFORMANCE

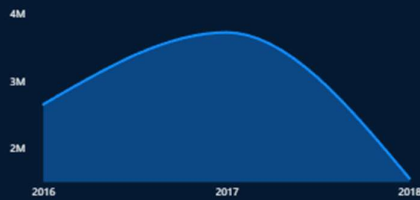
Total Revenue
85,78,491.31

Total order
1615

Active Customers
1445

Average Order Value
5,311.76

Sales by year



Sales by state



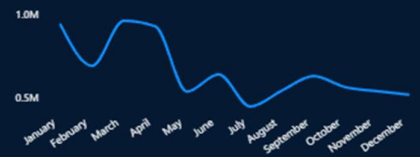
Order status

- ☐ Delivered
- ☐ Pending
- ☐ Rejected
- ☐ Shipped

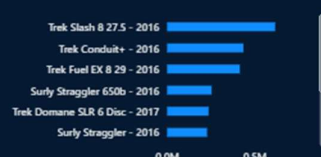
Store name

- ☐ Baldwin Bikes
- ☐ Rowlett Bikes
- ☐ Santa Cruz Bikes

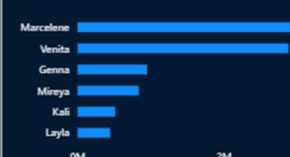
Monthly sale trend



Top Products by sales



Top customers by sales

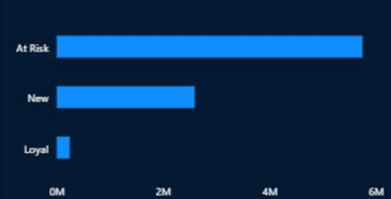


CUSTOMERS & PRODUCTS INSIGHTS

Product inventory

Product_name	Product_id	Quantity
Trek XM700+ Lowstep - 2018	188	27
Trek XM700+ Lowstep - 2018	188	29
Trek XM700+ Lowstep - 2018	188	30
Trek XM700+ - 2018	194	15
Trek XM700+ - 2018	194	20
Trek XM700+ - 2018	194	24
Trek X-Caliber Frameset - 2018	135	15
Trek X-Caliber Frameset - 2018	135	30
Trek X-Caliber 8 - 2018	118	9
Trek X-Caliber 8 - 2018	118	12
Trek X-Caliber 8 - 2018	118	25
Trek X-Caliber 8 - 2017	29	11
Trek X-Caliber 8 - 2017	29	13
Trek X-Caliber 8 - 2017	29	29
Trek X-Caliber 7 - 2018	129	10
Trek X-Caliber 7 - 2018	129	21
Trek Verve+ Lowstep - 2018	197	8
Trek Verve+ Lowstep - 2018	197	10
Trek Verve+ Lowstep - 2018	197	22
Total		

Sales by customer segment



low stock products
321

Total stock
13511

Order by City and status



Customers by segment



Power BI: DAX Calculations & Interactivity

Core DAX Measures

Total sales = sum('capstone order_items'[Total_price])

Average order value =Divide([Total_sales],Distinctcount('capstone orders'[order_id]))

low stock products = Calculate(Distinctcount('capstone products'[Product_id]),'capstone stocks'[Quantity]<10)

Total stock = Sum('capstone stocks'[Quantity])

Key Business Insights & Findings

1. Sales Trends Over Time

- Total sales reached the highest level in 2017 (over 3 Million).
- Every year, sales are very strong in March and April, reaching nearly 1.0 Million per month.
- After mid-year, sales remain steady at around 0.5 Million each month.

2. Customer Retention and Loyalty

- A large majority of customers (**81.18%**, or 1,170 people) are in the **At-Risk** group. This means they purchased once or twice earlier but have not returned recently.
- About **18.06%** are **New** customers.
- Only a small percentage are **Loyal** repeat buyers, showing that the store needs better customer follow-ups.

3. Best Selling Products

- Premium mountain bikes generate the highest total revenue.
- The top revenue-generating bikes are:
 - *Trek Slash 8 27.5 (2016)*
 - *Trek Conduit+ (2016)*
 - *Trek Fuel EX 8 29 (2016)*

4. Orders and Store Locations

- Cities like Mount Vernon, Scarsdale, and Ballston Spa placed the highest number of orders.
- Most customer orders are successfully delivered with very few cancellations or rejections.

Actionable Recommendations

1. Bring Back Inactive Customers

- Create special email and SMS campaigns for the **At-Risk** customer group (81.18%).
- Offer discount coupons, free service checks, or seasonal offers to bring them back to the store.

2. Smart Inventory Management

- Currently, **321 products** have very low stock (less than 10 units left).

3. Promote Popular Bikes and Add-ons

- Since Trek series bikes sell best, make sure they are always in stock in high-demand stores.
- Offer packages that bundle safety helmets, lights, and maintenance kits with high-end bikes to increase the Average Order Value (\$5,311.76).

4. Reward Top Staff and Stores

- Recognize and reward top-performing sales staff to keep motivation high.
- Give extra marketing support to top-performing branches like *Baldwin Bikes* to grow store sales further.

Conclusion & Project Deliverables

Summary of Deliverables

- **Excel Files:** Fully cleaned and standardized datasets saved as clean CSV files.
- **SQL Scripts:** Database schema, table structures, and analytical queries for sales, staff, and inventory.
- **Python Code:** Exploratory Data Analysis (EDA) and RFM customer segmentation code.
- **Power BI Dashboard:** An interactive report file (.pbix) with live KPI cards, visual charts, and filter slicers.
- **Final Report:** A structured 14-page document covering each tool and its findings.

Conclusion

- This project successfully combined **Excel, SQL, Python, and Power BI** to create a complete analytics workflow.
- The retail business can now easily track daily revenue, identify low stock immediately, and target specific customer groups with the right marketing strategies.
- This solution helps store managers make data-backed decisions quickly and with high confidence.